

Enhancing Digital Government & Economy (EDGE) Project
Institute of Information & Communication Technology
Dhaka University of Engineering & Technology, Gazipur
Digital Skills for Students Program

Applied Machine Learning — Final Project

Project 01

Student Dropout Risk Predictor

Tabular

Intermediate

Education

1. Project Overview

Student dropout is one of the most costly challenges facing educational institutions. Early identification of at-risk students allows advisors to intervene before it is too late. In this project, trainees will build a machine learning pipeline that analyses student academic, demographic, and socioeconomic features to predict the probability of dropout, and deliver an explanation of the top contributing risk factors through an interactive GUI.

Project Number:	Project 01
Data Type:	Tabular
Difficulty Level:	Intermediate
Domain:	Education
Submission Format:	Working code (GitHub), final report (PDF), live GUI demo

2. Learning Objectives

- Perform exploratory data analysis (EDA) to identify key predictors of student dropout.
- Handle missing values, encode categorical variables, and scale features appropriately.
- Train and compare at least three classifiers: Logistic Regression, Random Forest, and Decision Tree.
- Evaluate models using Accuracy, Precision, Recall, F1, and AUC-ROC.
- Build a GUI where an advisor enters student details and receives a dropout risk score with an explanation.

3. Dataset & Features

Recommended Datasets:

- UCI Student Performance / Dropout Dataset: <https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>
- Open University Learning Analytics (OULAD): https://analyse.kmi.open.ac.uk/open_dataset
- Kaggle — Students Performance Dataset: <https://www.kaggle.com/datasets/aljarah/xAPI-Edu-Data>

Key Features / Columns:

- Age at enrollment, marital status, nationality
- Previous qualification grade, admission grade
- Curricular units credited / enrolled / approved / grades
- Scholarship holder, tuition fees up to date, debtor status
- Target: Dropout / Enrolled / Graduate (binary: Dropout vs Non-dropout)

4. ML Pipeline & Implementation Steps

Step 1: Data Loading & EDA

Load CSV into pandas. Plot class distribution, correlation heatmap, boxplots of key features by class.

Step 2: Preprocessing

Impute missing values (median for numeric, mode for categorical). One-hot encode categoricals. Apply StandardScaler to numeric features.

Step 3: Model Training

Train Logistic Regression, Decision Tree, Random Forest. Use StratifiedKFold (5-fold) for cross-validation. Tune max_depth and n_estimators.

Step 4: Evaluation

Report classification report, confusion matrix, AUC-ROC curve. Plot feature importances from Random Forest.

Step 5: GUI

Form with student attribute inputs. Output: risk percentage, risk level badge (Low/Medium/High), top-3 contributing features highlighted.

5. GUI Requirement

Every project must include a functional GUI. Acceptable frameworks:

- Streamlit (recommended — easiest deployment)
- Tkinter (built-in Python — no installation required)
- Gradio (excellent for quick ML demos)

Minimum GUI requirements:

- Input interface (form fields, upload button, or text box as appropriate)
- Clear prediction output with confidence score
- At least one visualisation (chart, heatmap, or highlighted output)
- A disclaimer or limitation note visible to the user

6. Project Timeline & Milestones

Phase	Task / Deliverable	Target Week
Phase 1	Data collection, EDA report, class distribution, correlation heatmap	Day 1
Phase 2	Preprocessing pipeline: imputation, encoding, scaling	Day 2

Phase 3	Model training and 5-fold CV comparison table	Day 3
Phase 4	Evaluation: confusion matrix, AUC-ROC, feature importance	Day 3
Phase 5	GUI development, final report, live demo	Day 4 – Day 6

7. Evaluation Criteria

Criterion	Description	Marks
EDA & Preprocessing	Thoroughness of analysis, handling of missing data and encoding	20
Model Development	Correct implementation, comparison of 3+ models	25
Evaluation Metrics	Correct use of AUC-ROC, F1, confusion matrix	20
GUI Implementation	Functional, user-friendly interface with risk explanation	20
Report & Presentation	Clarity, structure, insights, demo quality	15
TOTAL		100

8. Tips & Guidance

- Use SMOTE or `class_weight='balanced'` to address class imbalance in the target.
- Display SHAP values or feature importances in the GUI to satisfy the explainability requirement.
- The Kaggle dataset is the most beginner-friendly starting point.
- If using Streamlit, deploy on Streamlit Cloud for a public demo link.

9. Final Deliverables

- GitHub Repository — clean, well-commented code with README.md
- Jupyter Notebook — EDA, preprocessing, model training, and evaluation
- Working GUI — demonstrated live during final presentation
- Final Report (PDF) — 8-12 pages covering all pipeline stages
- Presentation (5 minutes) — live demo + Q&A

Report must include: Title page, Abstract, Introduction, Dataset Description, Preprocessing, Model Architecture, Results & Evaluation, Limitations, Conclusion, References.